Assessment 3 Queries

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| Having queries regarding your assignment 3? Post them below

A3 FAQs

Students are expected to attempt the assessment independently. Teaching staff may help clarify what you need to do, but do not help on "how to" --- it is your job to work out "how to" based on what you have learned.

1. Sometimes I am getting an error through the streaming log. a FileNotFoundException when I put a new file to hdfs.

24/10/21 23:17:39 ERROR Executor: Exception in task 0.0 in stage 48.0 (TID 24)
java.io.FileNotFoundException: File does not exist: /Input/text1.txt._COPYING_
at org.apache.hadoop.hdfs.server.namenode.INodeFile.valueOf(INodeFile.java:72)
at org.apache.hadoop.hdfs.server.namenode.INodeFile.valueOf(INodeFile.java:62)
at org.apache.hadoop.hdfs.server.namenode.FSDirStatAndListingOp.getBlockLocations(FSDirStatAndL
istingOp.java:150)
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.getBlockLocations(FSNamesystem.java:1864
)
at org.apache.hadoop.hdfs.server.namenode.NameNodeRpcServer.getBlockLocations(NameNodeRpcServer
.java:743)
....
When this happens, the code stops working. If I run the code 10 times, it happens only once, and all other times I get the correct output. It seems this occurs when Spark attempts to read a file before it has been fully written, ultimately crashing the process. Does this mean there is an issue with my code?

hadoop fs -copyfromlocal creates _COPYING_. It's really luck based if the batch read happens while the file is uploading it'll crash and spark can't recover from checkpoint since the tmp file no longer exists.
COPYING is a temporary state file that is deleted after the upload. So if the program detects it as a new file, it starts glitching because the file is deleted ASAP after the upload.

Usually, the fix is to restart the program, clear input folder and try again and never rush to transfer your input files, let the program boot and warm up first (~20 seconds). then you can upload the files (small text file, about 5~10 lines) to the monitored dir every 5~10 seconds :-)

2. In the downloaded querylog_001.csv and querylog_002.csv, both contain header on the first line. However, in the assignment specification and appendix, both input query log files do NOT have header. May I double confirm if header will be included in the input file?

The samples provided in the assignment specification and appendix are intended to illustrate what each field looks like and give you a general understanding of the input format and the expected statistics collected.

For the actual assessment, the downloaded querylog_001.csv and querylog_002.csv files DO contain headers, and these are the input files your Spark Streaming job needs to handle. You need to correctly and appropriately handle the header rows in your code so that they are not processed as normal query log records.

Quick reminder on references:

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If a link appears in the course materials (lectures, slides, labs, Canvas pages), no separate reference is needed.

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If you use any resource not covered in the course (websites, textbooks, blogs, forums, sample

code, AI tools, etc.), you must include a reference.

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In line with RMIT Academic Integrity and the RMIT Referencing Guide: "You must acknowledge all the sources of information you have used in your assignments and research papers using an appropriate referencing style."

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For this course we're flexible about style (e.g., APA/IEEE/short form is fine), but you still need to reference any external resources not included in the course materials.

How to reference (short form is fine):

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Add a References section in your README/report, and/or cite inline near borrowed code.

- References
- Apache Hadoop, "HDFS Shell ('hdfs dfs') Command Guide," accessed 29 Sep 2025.

- When in doubt, cite it. This keeps things transparent and aligned with academic integrity.

Referencing isn't a course-specific rule; it's a university requirement you must follow throughout your studies.

[Luke George, 2026-10-21] Hi Ying,

For task 2 of this assignment, we need to find the number of the same query that was slow.

If I therefore theoretically had in a batch a query which contains "a" 5 times, with 1 of them "CRITICAL" slow and 2 of them "WARNING" slow (2 not slow), are we counting the number of the same query of the same level of slowness (1 Critical and 2 Warning) or the total of the same query which are slow (3 in this example) or just the overall amount of times the query appears? (5 overall).

I'm not 100% sure which way to go here, but I think we are counting based on the slowness level and the query (i.e: "a" and "WARNING" = 2)?

Could you please clarify this?

Thanks!
[Ying Zhang, 2026-10-21] Hi Luke,

slow_query_count counts all slow occurrences of the cleaned query, regardless of whether an individual occurrence might fall into a WARNING or CRITICAL range. The final severity_level is assigned once for the aggregated query, based on its maximum response time.

For example, given the input, With a slow-query threshold of 250 ms:

q011,u010,Hadoop MapReduce example,2026-05-01T09:00:50Z,desktop,0,350
q015,u010,Hadoop MapReduce example,2026-05-01T09:00:57Z,desktop,1,720

You should NOT produce separate records such as:

hadoop mapreduce example, WARNING, 1
hadoop mapreduce example, CRITICAL, 1

Instead, the required result is one aggregated record:

hadoop mapreduce example,2,720,535.00,1,CRITICAL

Please refer to the Task 2 example output in the Appendix and compare it carefully with your interpretation. It shows how slow_query_count and severity_level are expected to be reported.

Please let me know exactly which part of the example output is causing confusion, and I can clarify that specific point.

Regards,
Ying

[Shaun Leong, 2026-10-06] I might not be able to make it to the lab this week, will all the information and tutorials regarding the assignment from the in-person lab today be available through the online resources? thanks.
[Ying Zhang, 2026-10-06] Yes, you can find the recording in Collaborate Ultra.

Regards,

Ying

[Theeraj Kumar Kanagaraj, 2026-10-06] Hi, in the updated Appendix A, Task 1-001 shows spark streaming scala,2 and spark streaming,1, but Task 3-001 shows spark streaming,2 and spark streaming scala,1. Since Task 3 uses the same cleaned queries, could you please confirm which Task 3 output is expected?

Thank you

[Chi Wah Leung, 2026-10-14] Hi Teaching team,

In the downloaded querylog_001.csv and querylog_002.csv, both contain header on the first line. However, in the assignment specification and appendix, both input query log files do NOT have header. May I double confirm if header will be included in the input file? Thanks!

csv file screen shot:

Assignment Specification PDF:
[Ying Zhang, 2026-10-14] Hi there,

The samples provided in the assignment specification and appendix are intended to illustrate what each field looks like and give you a general understanding of the input format and the expected statistics collected.

For the actual assessment, the downloaded querylog_001.csv and querylog_002.csv files DO contain headers, and these are the input files your Spark Streaming job needs to handle. You need to correctly and appropriately handle the header rows in your code so that they are not processed as normal query log records.

Regards,

Ying

[Michael Jagiello, 2026-10-16] Hi,

I just wanted to quickly ask how expensive are tasks such as sorting when using spark and is it important we be mindful of this cost (and spend time optimising it) for our assignment?

Thank you!!
[Michael Jagiello, 2026-10-16] Also which libraries (if any) are we allowed to use for this assignment?
[Ying Zhang, 2026-10-16] You should use the libraries provided by Spark/Scala that are appropriate for the task. In particular, this assignment requires you to process the data using Spark Streaming DStreams, so your solution should be based on the DStream API and standard Spark/Scala functionality.

You should not replace the required Spark Streaming processing with a different framework or approach.

Regards,

Ying

[Ying Zhang, 2026-10-16] Hi Michael,

You are right that sorting (for example, sortByKey() which is a wide transformation) in Spark can be relatively expensive because it usually requires a shuffle, meaning data may need to be redistributed across partitions and transferred between executors.

For this assignment, you should be mindful of unnecessary expensive operations, but you do not need to spend excessive time trying to optimise every sort. If sorting is required by the specification, use the appropriate Spark operation and focus on producing the correct result with a sensible distributed design.

So the main principle is: avoid unnecessary shuffles, but do not remove operations that are required to meet the assignment specification.

Good luck,

Ying

[Luke George, 2026-10-19] Hi Ying,

I've noticed that both csv files are different, as one has an empty row at the bottom whereas the other does not.

This causes a bug with my code, and I am wondering whether the csv files are gmeant to be exactly the same structure or not?

($ at the bottom means extra line, which I think is a mistake)
[Ying Zhang, 2026-10-19] Hi Luke,

The empty row should be handled correctly in your code. You may assume that each valid data row contains the predefined columns specified in the assignment; however, your implementation should identify and ignore empty lines. Empty lines should always be handled appropriately in a streaming application.

Regards,

Ying

[Haasini Anand, 2026-10-20] Hello,

I was wondering if we have free reign over this assignment in terms of using UDF's (org.apache.spark.sql.functions.udf) and adjusting the dependencies to fit how we complete the task. Because as far as I know the criteria is only marking us based of the output and making sure the code that we write makes sense. Could you please clarify if we are allowed to parse the CSV input contents directly into a Spark DataFrame (using standard Spark reading methods) to complete the solution?

Thanks,
Haasini
[Ying Zhang, 2026-10-20] Hi Haasini,

You need to use the concepts, APIs, and techniques covered in this course to implement your solution for the given task.

This assignment is not assessed only on whether the final output is correct. Your implementation approach also needs to demonstrate your understanding of the course material.

For this task, you are expected to use Spark Streaming with DStreams, as taught in the course. Please base your solution on what you have learned in this course and follow the assignment specification carefully.

Regards,

Ying

[Bhanu Dhanwani, 2026-10-21] Hi Ying,

Could I please clarify the output directory numbering for batches with no qualifying records?

For example, if Task 2 generates task2-001 for the first batch, but the next batch contains no slow queries and produces no output, should the following non-empty Task 2 result be saved as task2-002 or task2-003?

I understand that empty RDDs should not produce output; I just want to confirm how the sequence numbers should be assigned.

Thank you!

[Ying Zhang, 2026-10-21] Hi Bhanu,

I think you may be mixing up two different cases, according to the spec: "For each RDD of the DStream, identify slow queries and produce a summary for each cleaned query in the current batch."

An empty RDD means there are no records in that batch at all.

A batch with 0 slow queries identified is different: the RDD may contain normal query records, but none of them meet the slow-query condition.

So:

- Empty RDD: no input records in that batch.

- Non-empty RDD with 0 slow queries: input records exist, but none are classified as slow.

These two cases should not be treated as the same thing in your implementation :-)

I'll leave the rest for you to work through based on the specification.

Regards,

Ying

[Haasini Anand, 2026-10-21] Hello,

I was wondering if I could get some help in terms of saving the files to a hdfs directory. As far as I am aware the code would process the certain number of lines from the csv files as singular lines (meaning it isnt concerned with what came before or after). When we want to save the output do we establish some sort of counter variable that will keep track of the batch?

If not, could you please point me in the right direction of where this example code might be? Because I have checked the class code starting from when we first introduced Spark, and I cant seem to find any place where we saved the contents to HDFS.

Thanks,
Haasini

[Ying Zhang, 2026-10-21] Hi Haasini,

We discussed how to write output to HDFS in the Week 9 practical sessions, so please review those materials again.

Regarding your question:

"When we want to save the output, do we establish some sort of counter variable that will keep track of the batch?"

This is exactly one of the implementation details you are expected to work out as part of the assignment.

The specification tells you what output is required for each batch and where it needs to be stored. You should use what you have learned about Spark Streaming, DStreams, and HDFS output to design an appropriate solution.

Regards,

Ying

[Luke George, 2026-10-24] Hi Ying,

I am a bit confused on what "output" means in this assignment.

"Only non-empty RDDs should generate output." - Does this mean an empty RDD isn't stored in output on HDFS, or prevent console printing? (for tasks 1 and 2)

Also, the UG requirement means every 3 seconds, we have to "output the accumulated query statistics, regardless of whether new query log files have arrived in the monitored HDFS directory". Does this mean store in HDFS every 3 seconds or just print every 3 seconds on to the console?

Thanks!

[Ying Zhang, 2026-10-24] Hi Luke,

For Tasks 1 and 2, "Only non-empty RDDs should generate output" refers to the required HDFS output. If the RDD for a batch is empty, you should not create an output folder/file for that batch. Console printing is not part of the required output.

For the UG Task 3 requirement, "output the accumulated query statistics every 3 seconds" also means save the required output to HDFS every 3-second batch, even if no new query log files arrive during that interval. It does not mean just printing the result to the console.

Regards,

Ying

[Luke George, 2026-10-25] Ok thanks!